



Modelling Financial Markets

Seminar 12: Options and the Volatility Surface — Hands-on



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Acronyms used in this material

- ▣ **AI:** Artificial Intelligence
- ▣ **ATM:** At-The-Money
- ▣ **CI:** Confidence Interval
- ▣ **CRR:** Cox–Ross–Rubinstein (binomial tree)
- ▣ **DM:** Diebold–Mariano (test)
- ▣ **DVOL:** Deribit Volatility Index
- ▣ **GBM:** Gradient Boosting Machine
- ▣ **GMM:** Generalized Method of Moments
- ▣ **HAC:** Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ **HAR-RV:** Heterogeneous AutoRegressive model of Realised Volatility
- ▣ **MSE:** Mean Squared Error
- ▣ **MZ:** Mincer–Zarnowitz (regression)
- ▣ **OLS:** Ordinary Least Squares
- ▣ **OTM:** Out-of-The-Money
- ▣ **QLIKE:** Quasi-LIKElihood (loss function)
- ▣ **SVI:** Stochastic Volatility Inspired (parameterisation)
- ▣ **UTC:** Coordinated Universal Time (Temps Universel Coordonné)
- ▣ **VIX:** CBOE Volatility Index
- ▣ **VIX3M:** Cboe 3-Month Volatility Index
- ▣ **VRP:** Variance Risk Premium

Today's Question and Route

- ▣ Question of the seminar: **how much can we trust the numbers that option prices give us**: Greeks, implied volatility, a fitted smile and the variance risk premium?
- ▣ Route
 - ▶ What you need for today: primer slides on options, Black–Scholes, the Greeks, implied volatility, the smile and the VIX ([Chapter 12](#) comes after the seminar)
 - ▶ A5, A7 on paper: one identity for all Greeks; implied volatility by Newton
 - ▶ B1: hedging error by simulation; B5: the variance risk premium with Newey–West errors
 - ▶ C2: find the errors in an AI answer; B6 at home; B3 (an SVI smile for Bitcoin) after the lecture
- ▣ What you produce: the A answers, two figures (B1, B5) and one table per B exercise, the corrected C2 code, one interpretation sentence per exercise
- ▣ Notebook for today: [open the seminar notebook in Colab](#); each task names its notebook section
- ▣ Seminar: Prof. Daniel Traian Pele, danpele@ase.ro; Antoaneta Amza, PhD candidate, antoaneta.amza@csie.ase.ro

Exercise Map

Exercise	Question	Type	Model
A5	why are gamma and vega of one expiry proportional?	Solved	–
A7	which volatility reproduces a market price of 4.50?	Solved	–
B1	how fast does the hedging error shrink with rebalancing?	Solved	–
B5	is the S&P 500 variance risk premium significant and stable?	Solved	–
C2	what is wrong in an AI answer?	Proposed	A5, A7
B6 (home)	is the VIX an efficient forecast of volatility?	Proposed	B5
B3 (after the lecture)	how precisely does SVI describe the Bitcoin smile?	Solved	–
Extension packs	A1–A4, A6, A8, B2, B4, B7, B8, C1, C3 (optional, after the seminar)	both	named on each task

- **[Solved]**: full solution in the slides and in the notebook, a model to copy; **[Proposed]**: you solve it, the solution is discussed in class
- Part A: derivations on paper; Part B: estimation and inference on real data, each ending with an interpretation question; Part C: open questions and AI critique

Data Used

Series	Symbol, field	Period	Units	Used in
S&P 500	GSPC.INDX, close	1990–2026	index points	B2, B5, B6, C1, C3
VIX, VIX3M	VIX.INDX, VIX3M.INDX, close	1990 / 2007–2026	annualised %	B2, B5, B6, C1
Bitcoin	BTC-USD.CC, close	2021–2026	USD	B7, C1
Bitcoin options	Deribit option snapshot: strike, type, bid, ask, mark price (BTC), mark IV, forward, index	22 Sep 2026, 07:51 UTC	BTC; % for IV	A1, A2, B3, B4, B8
DVOL	Deribit volatility index, daily	2021–2026	annualised %	B7, C1
Simulated	GBM paths, seed 42	–	–	B1

- Daily prices of the course data, one series per asset (field used: close), up to 18 September 2026; the Bitcoin options are a snapshot of Deribit quotes (the table of the chapter's Quantlet)
- Two series together: join the prices on common days first, then compute returns; C3 also needs the Kenneth French daily risk-free rate and the replication package of the case study

Setup: Run and Verify

```
# notebook section "Data" defines
# load_close(name) and deribit_chain()
px = pd.concat([load_close('sp500'),
                load_close('vix')],
               axis=1, join='inner').dropna()
print(len(px), px.index[0], px.index[-1])
print(px.iloc[[0, -1]])
c = deribit_chain()
print(len(c), c['expiry'].nunique())
```

- Expected output: 9 245 common days, 1990-01-02 to 2026-09-18

date	S&P 500	VIX
1990-01-02	359.69	17.24
2026-09-18	7 650.50	14.81

- Option snapshot: 986 contracts (calls and puts), 11 expiries
- If your counts differ, stop and check the symbols, the join and the end date

Notation Used in Every Exercise (1/2): Prices, Strikes and Time

- S : price of the underlying today; S_T : its price at expiry T ; K : strike (exercise price)
- $\tau = T - t$: time to expiry in years (90 calendar days: $\tau = 90/365$); r : continuously compounded risk-free rate
- σ : volatility, the annualised standard deviation of log returns (a decimal in formulas, in % when quoted)
- Forward price and log-moneyness

$$F = Se^{r\tau}, \quad k = \ln(K/F)$$

► $k < 0$: strike below the forward (OTM puts); $k > 0$: above it (OTM calls)

- σ_{imp} : implied volatility; total implied variance $w = \sigma_{imp}^2 \tau$
- Φ, φ : distribution function and density of the standard Normal distribution, $\varphi(x) = e^{-x^2/2}/\sqrt{2\pi}$
- "Vol points": percentage points of volatility; vega per vol point = vega per unit of σ divided by 100

Notation Used in Every Exercise (2/2): Black–Scholes and a Worked Line

- Black–Scholes call price [Black & Scholes \(1973\)](#) (no dividends)

$$C = S\Phi(d_1) - Ke^{-r\tau}\Phi(d_2), \quad d_{1,2} = \frac{\ln(S/K) + (r \pm \sigma^2/2)\tau}{\sigma\sqrt{\tau}}$$

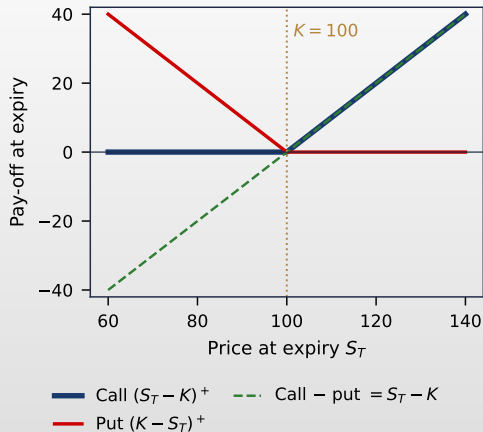
► $d_1 - d_2 = \sigma\sqrt{\tau}$; the put follows from put–call parity: $P = C - S + Ke^{-r\tau}$

- Greeks: $\Delta = \partial C / \partial S$, $\Gamma = \partial^2 C / \partial S^2$, vega $\mathcal{V} = \partial C / \partial \sigma$ per unit of σ
- One worked line: $S = K = 100$, $\tau = 0.5$, $r = 3\%$, $\sigma = 25\%$
 - $\sigma\sqrt{\tau} = 0.1768$, $d_1 = 0.1732$, $d_2 = -0.0035$, $C = 7.76$
- Daily log return in %: $r_t = 100 \ln(P_t/P_{t-1})$, P_t the close of day t ; A : days per year (252 trading or 365 calendar days)
- Realised variance over the next h days, annualised, in %²

$$RV_{t,t+h} = \frac{A}{h} \sum_{i=1}^h r_{t+i}^2$$

► $\sqrt{RV_{t,t+h}}$: realised volatility in % per year, comparable with the VIX [Carr & Wu \(2009\)](#)

What You Need for Today (1/18): Options and Their Pay-offs



- Chapter 12 comes after this seminar: these slides are the minimum for today's tasks
- **European call:** the right to buy at K on the expiry date; pay-off $(S_T - K)^+$, $x^+ = \max(x, 0)$
- **European put:** the right to sell at K ; pay-off $(K - S_T)^+$
- Moneyness
 - ▶ at the money (ATM): $K \approx S$
 - ▶ out of the money (OTM): a call with $K > S$, a put with $K < S$; zero pay-off if exercised today
- Green line: a long call and a short put pay $S_T - K$ in every state, like a forward bought at K

What You Need for Today (2/18): Put–Call Parity and No-Arbitrage Bounds

- **Put–call parity** (no dividends; C , P : today's prices of a call and a put with the same K and τ)

$$C - P = S - Ke^{-r\tau}$$

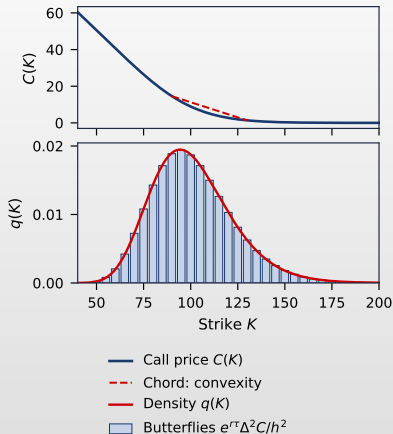
- ▶ $Ke^{-r\tau}$: today's value of K paid at expiry
- ▶ a call plus cash $Ke^{-r\tau}$ and a put plus one share both pay $\max(S_T, K)$, so they cost the same; no model is needed
- ▶ with $D = e^{-r\tau}$ and $F = Se^{r\tau}$: $C - P = DF - DK$, a straight line in K (A2)

- **No-arbitrage bounds**

$$(S - Ke^{-r\tau})^+ \leq C \leq S, \quad (Ke^{-r\tau} - S)^+ \leq P \leq Ke^{-r\tau}$$

- ▶ a price outside them is a data error or an arbitrage: a portfolio that costs nothing and never loses
- **Example**: $S = K = 100$, $r = 4\%$, $\tau = 0.5$: $Ke^{-r\tau} = 98.02$
 - ▶ a call at 7.00 implies the put $7.00 - 100 + 98.02 = 5.02$

What You Need for Today (3/18): Convexity in the Strike and the Risk-Neutral Density



- **Risk-neutral density** q of S_T : prices are discounted expectations under q

$$C(K) = e^{-r\tau} \int_K^\infty (x - K) q(x) dx$$

- Differentiating twice in K [Breedon & Litzenberger \(1978\)](#)

$$q(K) = e^{r\tau} \frac{\partial^2 C}{\partial K^2}$$

- **Butterfly** of step h : buy the calls $K - h$ and $K + h$, sell two calls K

$$C(K - h) - 2C(K) + C(K + h) \geq 0$$

- ▶ divided by h^2 , it approximates $e^{-r\tau} q(K)$ (bars)
- ▶ a negative butterfly is a negative density: an arbitrage or a quote error (A1, B8)

- Chart: Black-Scholes calls, $S = 100$, $\tau = 0.5$, $\sigma = 30\%$, $h = 5$

What You Need for Today (4/18): Geometric Brownian Motion and the Black–Scholes Price

- The stock as a geometric Brownian motion (GBM, [Chapter 11](#))

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

- ▶ μ : drift (expected return per year); W_t : standard Brownian motion, $W_t - W_s \sim N(0, t - s)$
- ▶ the log return over Δ years is $N((\mu - \sigma^2/2)\Delta, \sigma^2\Delta)$, so $S_t > 0$

- **Black–Scholes** [Black & Scholes \(1973\)](#): constant σ and r , continuous trading without costs

$$C = S\Phi(d_1) - Ke^{-r\tau}\Phi(d_2), \quad P = Ke^{-r\tau}\Phi(-d_2) - S\Phi(-d_1)$$

- ▶ $\Phi(d_2)$: risk-neutral probability that the call ends in the money; $\Phi(d_1)$: shares held in the replicating portfolio
- ▶ μ does not appear: Δ shares and cash, rebalanced continuously, replicate the option whatever the drift
- ▶ the price rises with S , σ , τ and r and falls with K

What You Need for Today (5/18): The Greeks

- **Greeks:** sensitivities of the call price, Black–Scholes without dividends

$$\Delta = \frac{\partial C}{\partial S} = \Phi(d_1), \quad \Gamma = \frac{\partial^2 C}{\partial S^2} = \frac{\varphi(d_1)}{S\sigma\sqrt{\tau}}, \quad \mathcal{V} = \frac{\partial C}{\partial \sigma} = S\varphi(d_1)\sqrt{\tau}$$

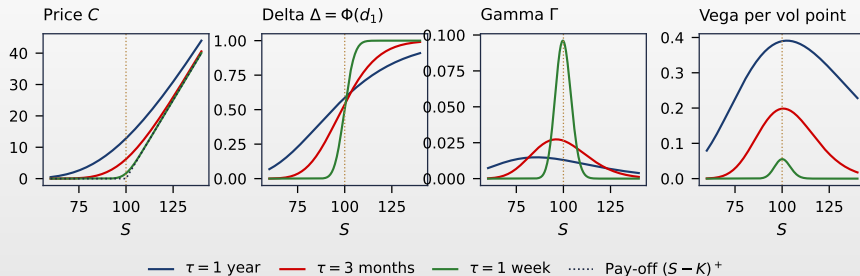
- ▶ delta $\Delta \in (0, 1)$: shares to hold per call; if S rises by 1, the call gains about Δ
- ▶ gamma $\Gamma > 0$: how fast delta moves; vega $\mathcal{V} > 0$: price change per unit of σ (per vol point: /100)
- ▶ theta $\Theta = \partial C / \partial t$, usually negative: the time value decays as expiry approaches

- **Second-order Taylor expansion** of the price change

$$\delta C \approx \Delta \delta S + \frac{1}{2} \Gamma (\delta S)^2 + \mathcal{V} \delta \sigma + \Theta \delta t$$

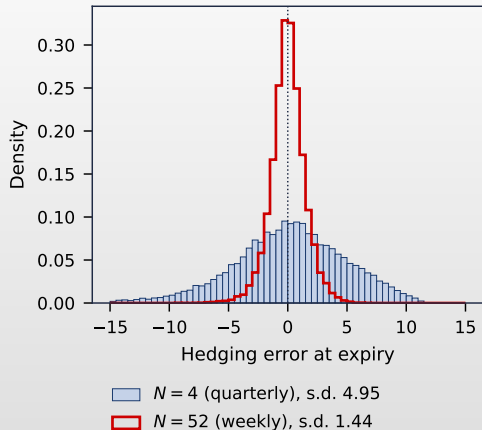
- **Portfolio Greeks add up:** n_i units of option i give $\Delta_{\Pi} = \sum_i n_i \Delta_i$, $\Gamma_{\Pi} = \sum_i n_i \Gamma_i$
 - ▶ delta–gamma hedge: a second option sets $\Gamma_{\Pi} = 0$, then shares ($\Delta = 1$, $\Gamma = 0$) set $\Delta_{\Pi} = 0$ (A6)

What You Need for Today (6/18): Price and Greeks as Functions of S



- Call with $K = 100$, $r = 2\%$, $\sigma = 30\%$, for three times to expiry τ
- Price: above the pay-off by the time value, which vanishes as $\tau \rightarrow 0$
- Delta: an S-shaped curve from 0 to 1, steeper near expiry
- Gamma and vega: largest near K ; gamma grows as expiry approaches, vega shrinks

What You Need for Today (7/18): Delta Hedging and the Hedging Error



□ Delta hedging a sold call

- ▶ receive C_0 , buy Δ_0 shares, keep $B_0 = C_0 - \Delta_0 S_0$ in cash at rate r
- ▶ at each of N dates reset Δ , paying from cash: the portfolio is self-financing

□ Hedging error at expiry

$$e_T = \Delta_{N-1} S_T + B_T - (S_T - K)^+$$

- ▶ B_T : cash at T , interest included; $e_T = 0$ with continuous rebalancing in the Black–Scholes model
- ▶ between dates gamma makes the hedge imperfect: $\text{s.d.}(e_T) \propto 1/\sqrt{N}$

- Chart (simulated): one-year ATM call, $\sigma = 30\%$; 13 times more trades cut the s.d. from 4.95 to 1.44, close to the factor $\sqrt{13} = 3.6$

What You Need for Today (8/18): Itô's Lemma, the Black–Scholes Equation, the Hedging P&L

- **Itô's lemma** ([Chapter 11](#)) for an option value $V(S, t)$ when $dS = \mu S dt + \sigma S dW$

$$dV = \left(V_t + \mu S V_S + \frac{1}{2} \sigma^2 S^2 V_{SS} \right) dt + \sigma S V_S dW_t$$

- ▶ V_t, V_S, V_{SS} : partial derivatives; the term $\frac{1}{2} \sigma^2 S^2 V_{SS}$ comes from $(dS)^2 = \sigma^2 S^2 dt$

- **Black–Scholes equation**: holding $\Delta = V_S$ removes dW and μ ; a riskless position earns r

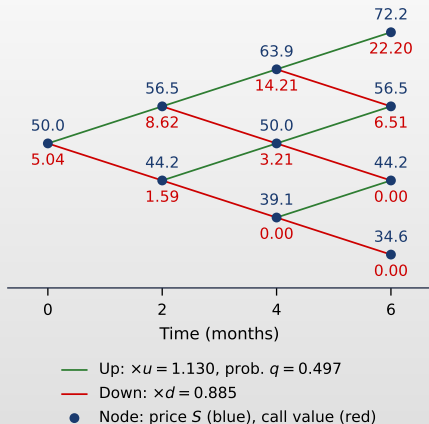
$$V_t + \frac{1}{2} \sigma^2 S^2 V_{SS} + r S V_S - r V = 0$$

- **Hedging P&L when volatility is wrong** (A3): sold at σ_{imp} , the price moves with σ_{real}

$$d \text{P\&L} = r \text{P\&L} dt + \frac{1}{2} \Gamma S^2 (\sigma_{imp}^2 - \sigma_{real}^2) dt$$

- ▶ the seller gains when realised volatility is below implied; the gain accrues through ΓS^2 , so it depends on the path

What You Need for Today (9/18): The Binomial Tree



- **Cox–Ross–Rubinstein** Cox, Ross & Rubinstein (1979): in each step of length $\Delta t = T/N$ the price moves to Su or Sd

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = \frac{1}{u}, \quad q = \frac{e^{r\Delta t} - d}{u - d}$$

- q : risk-neutral probability of an up move; the expected price grows at the rate r

- **Backward induction**: start from the pay-offs at T , then at each node

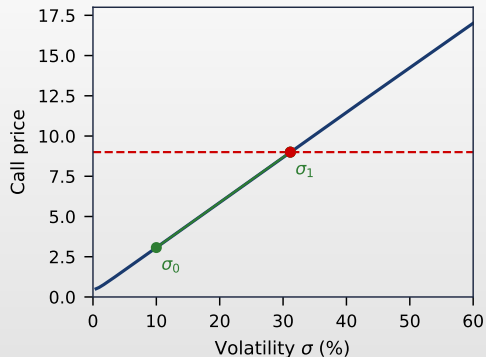
$$V = e^{-r\Delta t} [qV_u + (1 - q)V_d]$$

- V_u, V_d : option values at the two next nodes

- Chart: $S = K = 50$, six months, $r = 4\%$, $\sigma = 30\%$, $N = 3$: $C_0 = 5.04$

- As $N \rightarrow \infty$, C_N tends to the Black–Scholes price (A4)

What You Need for Today (10/18): Implied Volatility and Newton's Method



— $BS(\sigma)$ — Tangent (Newton step)
 - - - Market price $C^{mkt} = 9$

- S, K, τ, r are observed; σ is not. **Implied volatility** σ_{imp} solves

$$BS(\sigma_{imp}) = C^{mkt}$$

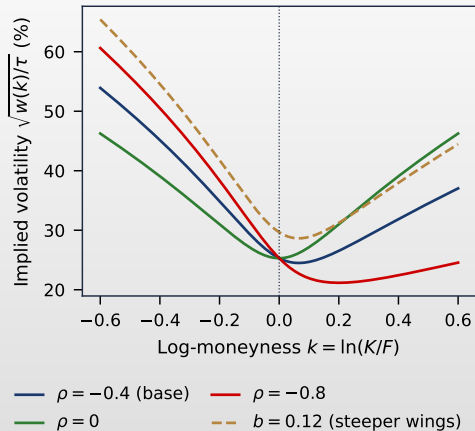
- ▶ $BS(\sigma)$: Black–Scholes price; C^{mkt} : market price
- ▶ unique, since vega > 0 ; it exists iff C^{mkt} lies strictly inside the no-arbitrage bounds

- **Newton's method**: follow the tangent from the iterate σ_k

$$\sigma_{k+1} = \sigma_k - \frac{BS(\sigma_k) - C^{mkt}}{\mathcal{V}(\sigma_k)}$$

- Chart: $S = K = 100, \tau = 0.5, r = 1\%, C^{mkt} = 9$: from $\sigma_0 = 10\%$ to $\sigma_1 = 31.14\%$, then 31.15%
- ATM rule of thumb [Brenner & Subrahmanyam \(1988\)](#): $C \approx 0.4 S \sigma \sqrt{\tau}$, a starting value
- Traders quote options in σ_{imp} : comparable across strikes, maturities and assets

What You Need for Today (11/18): The Smile and the SVI Curve



- If Black–Scholes were exact, σ_{imp} would not depend on K ; in the data it does: the **smile** (or skew)
- **SVI Gatheral & Jacquier (2014)**: total implied variance $w = \sigma_{imp}^2 \tau$ as a function of $k = \ln(K/F)$

$$w(k) = a + b \left(\rho(k - m) + \sqrt{(k - m)^2 + s^2} \right)$$

- ▶ a : level; $b \geq 0$: slope of the wings
- ▶ $\rho \in (-1, 1)$: tilt; $\rho < 0$ makes the put wing dearer (skew)
- ▶ m : horizontal shift; $s > 0$: curvature at the minimum
- ▶ in the wings w is linear in k , with slopes $b(1 \pm \rho)$
- Chart: $a = 0.004$, $b = 0.08$, $m = 0$, $s = 0.15$, $\tau = 0.25$; one parameter changed at a time

What You Need for Today (12/18): Fitting a Smile and Reading a Density from It

- **Least squares** on n quotes (k_i, w_i) , $\theta = (a, b, \rho, m, s)$

$$\hat{\theta} = \arg \min_{\theta} \sum_{i=1}^n (w(k_i; \theta) - w_i)^2$$

- ▶ fit quality: root mean squared error of $\sqrt{w/\tau}$, in vol points
- ▶ **pairs bootstrap**: resample the quotes with replacement, refit, read 2.5% and 97.5% percentiles [Efron \(1979\)](#)

- **Butterfly condition** [Gatheral & Jacquier \(2014\)](#): the density is nonnegative iff $g(k) \geq 0$ for all k

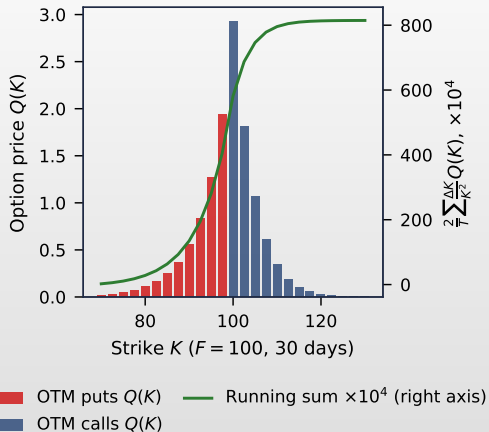
$$g(k) = \left(1 - \frac{k w'}{2w}\right)^2 - \frac{w'^2}{4} \left(\frac{1}{w} + \frac{1}{4}\right) + \frac{w''}{2}$$

- ▶ w' , w'' : first and second derivatives of $w(k)$

- **Black-76** prices from the fitted w (forward F , discount $e^{-r\tau}$), then $q(K) = e^{r\tau} \partial^2 C / \partial K^2$ (B8)

$$C = e^{-r\tau} [F\Phi(d_1) - K\Phi(d_2)], \quad d_{1,2} = \frac{-k \pm w/2}{\sqrt{w}}$$

What You Need for Today (13/18): Pricing Variance with Options: the VIX



- **Variance swap**: pays $RV - K_{var}$ at T ; its fair strike is a strip of OTM options [Bakshi, Kapadia & Madan \(2003\)](#)

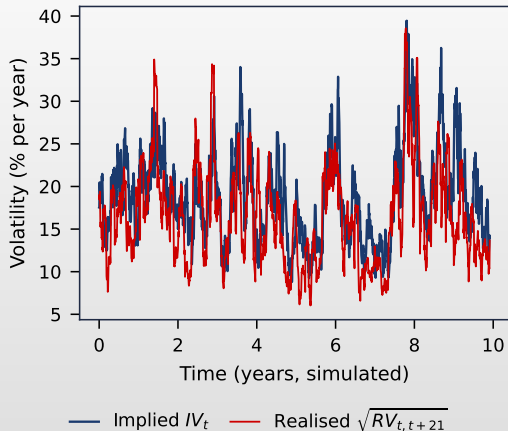
$$K_{var} = \frac{2e^{rT}}{T} \left[\int_0^F \frac{P(K)}{K^2} dK + \int_F^\infty \frac{C(K)}{K^2} dK \right]$$

- **Cboe formula** [Cboe \(2025\)](#) for a finite strip

$$\sigma^2 = \frac{2}{T} \sum_i \frac{\Delta K_i}{K_i^2} e^{rT} Q(K_i) - \frac{1}{T} \left(\frac{F}{K_0} - 1 \right)^2$$

- ▶ $Q(K_i)$: OTM mid price (puts below K_0 , calls above); K_0 : first strike at or below F ; ΔK_i : half the gap between neighbouring strikes
- ▶ $VIX = 100\sqrt{\sigma^2}$ interpolated to 30 days; VIX^2 : the risk-neutral expected variance, in $\%^2$ a year
- $VIX = 20$: about 20% a year, $20/\sqrt{12} \approx 5.8\%$ a month

What You Need for Today (14/18): The Variance Risk Premium



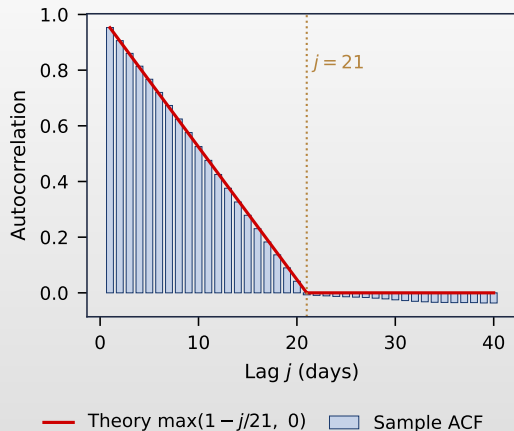
- **Variance risk premium Carr & Wu (2009):** implied minus subsequently realised variance

$$VRP_t = IV_t^2 - RV_{t,t+h}$$

- ▶ IV_t : implied volatility in % (VIX or DVOL); in vol points $IV_t - \sqrt{RV_{t,t+h}}$
- ▶ the profit of a seller of variance insurance; positive on average if investors pay to insure high-volatility states

- Example: $VIX = 20$, realised volatility 15%:
 $VRP = 400 - 225 = 175 \text{ \%}^2$, i.e. 5 vol points
- Chart (simulated): implied variance 20% above the expected variance; the premium is positive on 71% of days but negative in volatility spikes
- Variance swap with vega notional 1: P&L $\approx (IV^2 - RV)/(2 IV)$ vol points (C1)

What You Need for Today (15/18): Overlapping Windows and Newey–West Errors

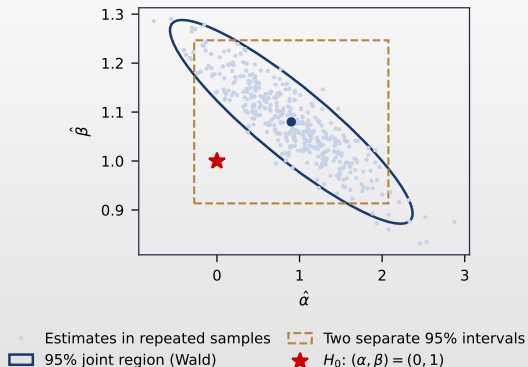


- Consecutive $RV_{t,t+21}$ share 20 of 21 days: even for i.i.d. returns their autocorrelation at lag j is $1 - j/21$ (chart)
- The naive s.e. $s/\sqrt{n}$ of a mean $\bar{x}$ is then too small; **Newey–West** [Newey & West \(1987\)](#) with L lags

$$\hat{\Omega} = \hat{\gamma}_0 + 2 \sum_{j=1}^L \left(1 - \frac{j}{L+1}\right) \hat{\gamma}_j, \quad \text{se}(\bar{x}) = \sqrt{\hat{\Omega}/n}$$

- ▶ $\hat{\gamma}_j$: sample autocovariance at lag j ; the Bartlett weights keep $\hat{\Omega} \geq 0$; take $L \geq h$
- ▶ $t = \bar{x}/\text{se}(\bar{x})$ is approximately $N(0, 1)$: $|t| > 1.96$ rejects H_0 : mean = 0 at 5%
- Two periods: regress $x_t = c_0 + c_1 D_t + u_t$, $D_t = 1$ after the break; HAC t -test of $c_1 = 0$
- Alternative: **block bootstrap** [Künsch \(1989\)](#), resampling blocks of consecutive days

What You Need for Today (16/18): Is a Forecast Unbiased? Mincer–Zarnowitz and Wald



- ▣ **Mincer–Zarnowitz regression** Mincer & Zarnowitz (1969) of the outcome on the forecast

$$RV_{t,t+21} = \alpha + \beta VIX_t^2 + \varepsilon_t$$

- ▶ unbiased forecast: $\alpha = 0$ and $\beta = 1$ jointly; efficient: no other variable known at t adds information

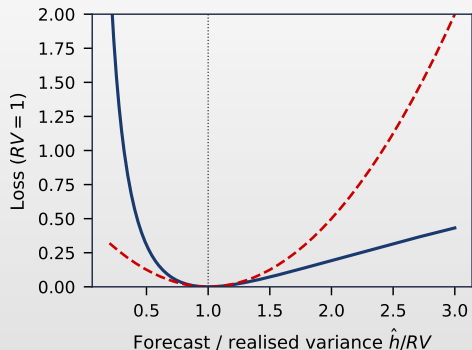
- ▣ **Wald test** of q linear restrictions $H_0: R\theta = c$

$$W = (R\hat{\theta} - c)^\top (R\hat{V}R^\top)^{-1} (R\hat{\theta} - c) \sim \chi_q^2$$

- ▶ $\hat{\theta} = (\hat{\alpha}, \hat{\beta})$; $\hat{V}$: HAC covariance; here $R = I_2$, $c = (0, 1)$, $q = 2$; reject at 5% if $W > 5.99$

- ▣ **Chart:** $\text{corr}(\hat{\alpha}, \hat{\beta}) = -0.9$; $t_\alpha = 1.50$, $t_\beta = 0.94$, yet $W = 29.9$: $(0, 1)$ is inside the box, outside the ellipse

What You Need for Today (17/18): Comparing Variance Forecasts: HAR-RV, QLIKE, Diebold–Mariano



- QLIKE: $RV/\hat{h} - \ln(RV/\hat{h}) - 1$
- - Squared error $(\hat{h} - RV)^2/2$

- **HAR-RV in logs** Corsi (2009): daily, weekly, monthly components

$$\ln RV_{t,t+21} = b_0 + \sum_{j \in \{1,5,22\}} b_j \ln RV_t^{(j)} + u_t$$

- ▶ $RV_t^{(j)}$: mean of the last j squared daily returns, annualised; level forecast $\exp(\widehat{\ln RV} + \hat{s}^2/2)$, $\hat{s}^2$: residual variance

- **QLIKE loss** Patton (2011) of a variance forecast $\hat{h}$

$$L(RV, \hat{h}) = \frac{RV}{\hat{h}} - \ln \frac{RV}{\hat{h}} - 1 \geq 0$$

- ▶ zero only if $\hat{h} = RV$; penalises under-prediction more (chart); robust to noise in RV

- **Diebold–Mariano** Diebold & Mariano (1995):
 $d_t = L_t^A - L_t^B$, $DM = \bar{d}/se_{HAC}(\bar{d}) \approx N(0, 1)$

What You Need for Today (18/18): Tools for the Extension Packs

- **Delta method** (A2): standard error of a smooth function $g(\hat{\theta})$ of estimates with covariance $\hat{V}$

$$\text{se}(g(\hat{\theta})) \approx \sqrt{\nabla g(\hat{\theta})^\top \hat{V} \nabla g(\hat{\theta})}$$

- **Instrumental variable** z_t (B6) Christensen & Prabhala (1998): correlated with the regressor, not with the error;
 $\hat{\beta}_{IV} = \widehat{\text{cov}}(z, y) / \widehat{\text{cov}}(z, x)$

- **Delta-hedged option gain** Bakshi & Kapadia (2003) (B2): option value change minus the hedge gains over $[t, t + \tau]$

$$\Pi_{t,t+\tau} = C_{t+\tau} - C_t - \sum_{n=0}^{N-1} \Delta_{t_n} (S_{t_{n+1}} - S_{t_n})$$

- ▶ mean zero if options are priced at the volatility that is realised; negative for the buyer if a variance premium exists

- **Martin bound** Martin (2017) (C3): a lower bound on the expected excess return of the market

$$E_t[R_{t,T}] - R_{f,t} \geq \frac{1}{R_{f,t}} \text{var}_t^*(R_{t,T}) = \frac{2}{S_t^2} \int_0^\infty \text{OTM}_t(K) dK$$

- ▶ $R_{t,T}$: gross market return; $R_{f,t}$: gross risk-free return; var^* : risk-neutral variance; $\text{OTM}_t(K)$: OTM option price at strike K

A5 [Solved]: One Identity for All the Greeks — Task

▣ Questions

- ▶ why are the gamma and the vega of options with the same expiry proportional?
- ▶ what are they for an at-the-money call?

▣ Inputs

- ▶ The Black–Scholes call of the notation slide; no dividends
- ▶ Numbers: $S = K = 100$, six months ($\tau = 0.5$), $r = 3\%$, $\sigma = 25\%$; no data file

▣ Tasks

1. Prove the identity $S\varphi(d_1) = Ke^{-r\tau}\varphi(d_2)$ by computing $d_1^2 - d_2^2$.
2. Use the identity to show that $\Delta = \Phi(d_1)$ and vega $= S\varphi(d_1)\sqrt{\tau}$.
3. Show that vega $= \Gamma S^2\sigma\tau$, with $\Gamma = \varphi(d_1)/(S\sigma\sqrt{\tau})$.
4. Compute d_1 , d_2 , both sides of the identity, Δ , Γ , vega in two ways, the call, and the put from put–call parity.

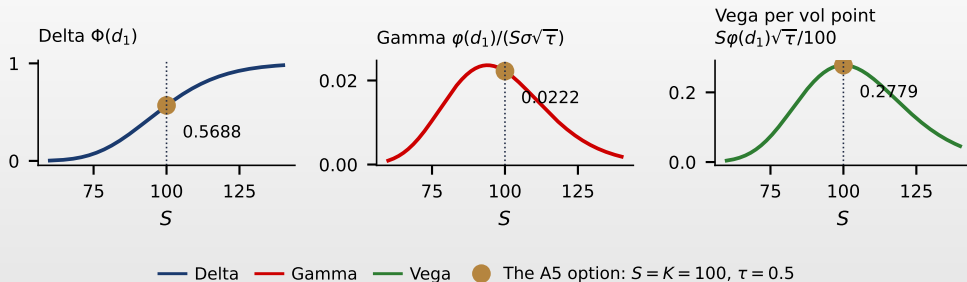
- ▣ **Report:** the three proofs, a table of the numbers, and one sentence on what vega $= \Gamma S^2\sigma\tau$ implies for hedging gamma with options of the same expiry

- ▣ Notebook: section A5

A5 [Solved]: Solution (1/2) — Proofs

- Step 1: $\varphi(d_2)/\varphi(d_1) = e^{(d_1^2 - d_2^2)/2}$ and $d_1^2 - d_2^2 = (d_1 - d_2)(d_1 + d_2) = \sigma\sqrt{\tau} \cdot \frac{2\ln(S/K) + 2r\tau}{\sigma\sqrt{\tau}}$
- Hence $\varphi(d_2)/\varphi(d_1) = Se^{r\tau}/K$, i.e. $S\varphi(d_1) = Ke^{-r\tau}\varphi(d_2)$
- Step 2: $\partial_S C = \Phi(d_1) + [S\varphi(d_1) - Ke^{-r\tau}\varphi(d_2)]\partial_S d_1 = \Phi(d_1)$, since $\partial_S d_1 = \partial_S d_2$ and the bracket is 0 by Step 1
- Vega: $S\varphi(d_1)\partial_\sigma d_1 - Ke^{-r\tau}\varphi(d_2)\partial_\sigma d_2 = S\varphi(d_1)\partial_\sigma(d_1 - d_2) = S\varphi(d_1)\sqrt{\tau}$
- Step 3: $\Gamma S^2\sigma\tau = \frac{\varphi(d_1)}{S\sigma\sqrt{\tau}} S^2\sigma\tau = S\varphi(d_1)\sqrt{\tau} = \text{vega}$

A5 [Solved]: Solution (2/2) — Numbers and Chart



- Step 4: $\sigma\sqrt{\tau} = 0.1768$, $d_1 = (0.03 + 0.03125) \cdot 0.5 / 0.1768 = 0.1732$, $d_2 = -0.0035$; identity: $100 \cdot 0.3930 = 39.30$ and $98.51 \cdot 0.3989 = 39.30$
- $\Delta = 0.5688$, $\Gamma = 0.0222$; vega = $100 \cdot 0.3930\sqrt{0.5} = 27.79 = \Gamma S^2 \sigma \tau = 27.79$ per unit of σ (0.2779 per vol point); $C = 7.76$, $P = C - 100 + 98.51 = 6.27$
- Interpretation:** for one expiry vega and gamma are proportional, so a gamma-neutral book of same-expiry options is also vega-neutral, but only while the smile is flat (extension A6)

A7 [Solved]: Implied Volatility by Newton's Method — Task

□ Questions

- ▶ which volatility makes the Black–Scholes price equal to a market price of 4.50?
- ▶ how fast does Newton's method find it?

- **Inputs:** a European call with $S = K = 100$, three months ($\tau = 0.25$), $r = 2\%$, market price 4.50; vega $\mathcal{V}(\sigma) = S\varphi(d_1)\sqrt{\tau}$ from A5

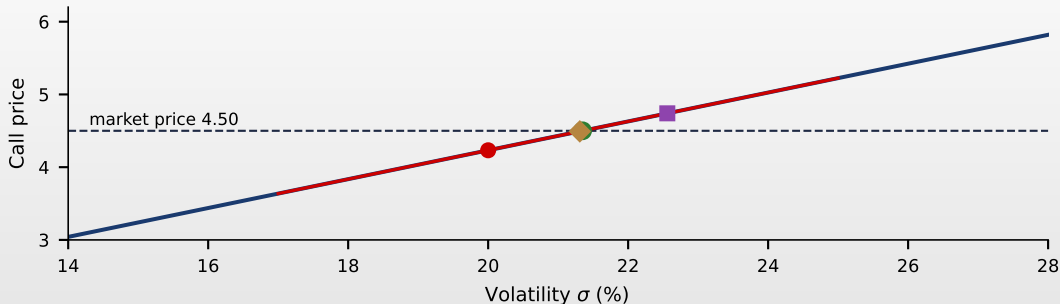
□ Tasks

1. Prove that $\sigma \mapsto BS(\sigma)$ is continuous and strictly increasing, with limits $(S - Ke^{-r\tau})^+$ as $\sigma \rightarrow 0$ and S as $\sigma \rightarrow \infty$, and state when an implied volatility exists.
 2. Derive the at-the-money-forward approximation $C \approx S\sigma\sqrt{\tau}/\sqrt{2\pi}$ [Brenner & Subrahmanyam \(1988\)](#).
 3. Invert it to get two starting values: from the price 4.50, and from 4.50 minus half the discounted forward intrinsic value $(S - Ke^{-r\tau})/2$.
 4. Run one Newton step $\sigma_1 = \sigma_0 - (BS(\sigma_0) - 4.50)/\mathcal{V}(\sigma_0)$ from $\sigma_0 = 20\%$ and compare σ_1 with the exact implied volatility.
- **Report:** the two starting values, σ_1 and the exact implied volatility; one sentence on why Newton converges so fast here and when it would fail
 - **Notebook:** section A7

A7 [Solved]: Solution (1/3) — Existence and Approximation

- Step 1: $\partial_\sigma BS = S\varphi(d_1)\sqrt{\tau} > 0$ (A5): strictly increasing and continuous
- $\sigma \rightarrow 0$: $d_{1,2} \rightarrow \pm\infty$ according to the sign of $\ln(S/K) + r\tau$, so $BS \rightarrow (S - Ke^{-r\tau})^+$; $\sigma \rightarrow \infty$: $d_1 \rightarrow \infty$, $d_2 \rightarrow -\infty$, so $BS \rightarrow S$
- By the intermediate value theorem a unique σ_{imp} exists if and only if the price lies strictly inside the no-arbitrage bounds $(S - Ke^{-r\tau})^+ < C < S$
- Step 2: if $S = Ke^{-r\tau}$, $d_{1,2} = \pm\sigma\sqrt{\tau}/2$ and $C = S[\Phi(\sigma\sqrt{\tau}/2) - \Phi(-\sigma\sqrt{\tau}/2)] \approx S\sigma\sqrt{\tau}\varphi(0) = S\sigma\sqrt{\tau}/\sqrt{2\pi}$

A7 [Solved]: Solution (2/3) — Newton Step and Chart

— Black-Scholes price $BS(\sigma)$ ● Newton $\sigma_1 = 21.35\%$ (exact 21.35%)

◆ Start with correction 21.31%

— Tangent at $\sigma_0 = 20\%$ (Newton step)

■ Start without correction 22.56%

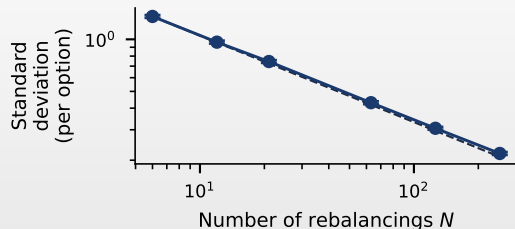
A7 [Solved]: Solution (3/3) — Newton Step and Chart

- Step 3: without correction $\sigma \approx 4.50\sqrt{2\pi}/(100 \cdot 0.5) = 22.56\%$
- With correction: $Ke^{-r\tau} = 99.501$, half the forward intrinsic value 0.249, $\sigma \approx (4.50 - 0.249)\sqrt{2\pi}/50 = 21.31\%$
- Step 4: $BS(20\%) = 4.2322$, $\mathcal{V} = 19.848$, so $\sigma_1 = 0.20 - (4.2322 - 4.50)/19.848 = 21.3495\%$; exact 21.3495%
- **Interpretation:** near the money the price is almost linear in σ , so one step gives four decimals; deep out of the money vega ≈ 0 and Newton can overshoot: use a bracketing solver (Brent)

B1 [Solved]: Hedging Error and Rebalancing Frequency — Task

- **Question:** how fast does the hedging error of a delta-hedged call shrink as the dealer rebalances more often?
- **Inputs**
 - ▶ Simulated data: 10 000 paths of geometric Brownian motion, $S_0 = 100$, $\mu = 8\%$, $\sigma = 20\%$, 252 steps over three months (seed 42)
 - ▶ A three-month ATM call ($K = 100$, $r = 4\%$) sold at its Black–Scholes premium 4.49 and delta-hedged at $\sigma = 20\%$
- **Tasks**
 1. For each $N \in \{6, 12, 21, 63, 126, 252\}$, rebalance the hedge N times and compute on every path the hedging error at expiry: shares plus cash minus the pay-off.
 2. Compute the standard deviation of the error for each N and a 95% percentile bootstrap interval from 1 000 resamples of the paths [Efron \(1979\)](#).
 3. Regress $\ln(\text{s.d.})$ on $\ln N$ and give the slope with a bootstrap interval from the same resamples.
 4. Find the smallest N in the list for which the standard deviation is below 10% of the premium.
- **Report:** a table of standard deviations with intervals, the log-log chart and the slope; one sentence on whether the slope is consistent with $-1/2$ and what transaction costs would change
- **Notebook:** section B1

B1 [Solved]: Solution (1/2) — Chart and Table



--- Slope $-1/2$ — Standard deviation of the hedging error, 95% bootstrap interval

- Step 1: on each path, sell the call at 4.49, buy Δ_0 shares, finance with cash at r ; at each rebalancing date reset Δ ; at expiry, error = shares + cash — pay-off
- Step 2: the s.d. falls from 1.356 ($N = 6$) to 0.219 ($N = 252$); the intervals are narrow because 10 000 paths give a precise s.d.

N	6	12	21	63	126	252
s.d.	1.356	0.963	0.744	0.430	0.306	0.219
95% CI	[1.331, 1.381]	[0.945, 0.983]	[0.731, 0.758]	[0.422, 0.438]	[0.300, 0.312]	[0.215, 0.223]

B1 [Solved]: Solution (2/2) — Slope and Interpretation

- Step 3: slope -0.489 , 95% CI $[-0.496, -0.482]$: close to $-1/2$ and slightly above it at these finite N ; the $1/\sqrt{N}$ law of the primer is asymptotic
- Step 4: daily hedging ($N = 63$) leaves 9.6% of the premium, about the 10% target; four times a day halves it again (ratio 1.97)
- **Interpretation:** with a fixed fee per trade, costs grow like N while the error falls like $1/\sqrt{N}$; the best frequency depends on the cost model and on the trade-off chosen between risk and cost

B5 [Solved]: The Variance Risk Premium of the S&P 500 — Task

▣ Questions

- ▶ is the variance risk premium of the S&P 500 significantly positive?
- ▶ was it the same before and after 2008?

▣ Inputs

- ▶ Daily S&P 500 and VIX closes joined on common days, January 1990 – August 2026 (the last 21-day window ends in September 2026)
- ▶ $RV_{t,t+21} = \frac{252}{21} \sum_{i=1}^{21} r_{t+i}^2$ in %²; $VRP_t = VIX_t^2 - RV_{t,t+21}$; in vol points $VIX_t - \sqrt{RV_{t,t+21}}$ Carr & Wu (2009)

▣ Tasks

1. Build $RV_{t,t+21}$ from the next 21 log returns and VRP_t on every day with a complete window.
2. Compute the mean VRP, its Newey–West standard error with 21 lags Newey & West (1987), a 95% interval, and the naive standard error $s/\sqrt{n}$.
3. Repeat step 2 in vol points and compute the share of days with a positive premium.
4. Repeat for 1990–2007 and 2008–2026, then test equality by regressing VRP_t on a constant and a 2008–2026 dummy with Newey–West errors (21 lags).

▣ Report: a table by period and the chart; one sentence each

- ▶ why is the naive standard error wrong here?
- ▶ by how much is it wrong?
- ▶ is the premium stable?
- ▶ why is the variance premium less precise than the vol-point premium?

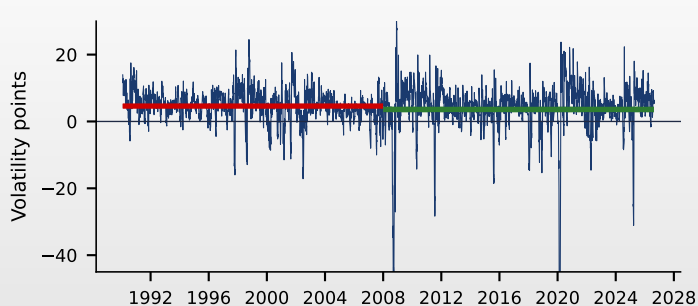
▣ Notebook: section B5

B5 [Solved]: Solution (1/3) — Table and Steps

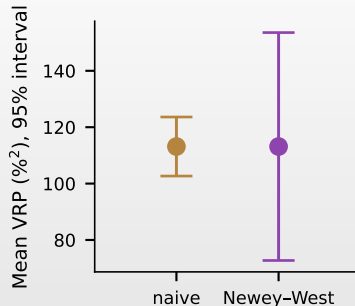
	n	Mean VRP ($\%^2$)	95% CI	Vol points (s.e.)	Positive
1990–2026	9 203	113.1	[72.7, 153.6]	4.09 (0.27)	86%
1990–2007	4 516	149.3	[125.2, 173.5]	4.60 (0.25)	88%
2008–2026	4 687	78.3	[2.7, 153.9]	3.60 (0.46)	84%

- Step 1: $RV_{t,t+21}$ from the next 21 log returns, then VRP_t on every day with a complete window
- Step 2: regress VRP_t on a constant with HAC standard errors (Bartlett, 21 lags: the overlap); interval = mean ± 1.96 s.e.; full sample $t = 113.1/20.6 = 5.49$: reject H_0 at 5%
- Naive s.e. 5.3 vs Newey–West 20.6: 3.9 times too small, because consecutive windows share 20 of 21 days

B5 [Solved]: Solution (2/3) — Chart and Interpretation



- Daily premium $VIX_t - \sqrt{RV_{t,t+21}}$
- Mean 1990-2007: 4.60 (NW 95%: ± 0.50)
- Mean 2008-2026: 3.60 (NW 95%: ± 0.90)



- Naive $s/\sqrt{n}$ interval
- Newey-West (21 lags) interval

B5 [Solved]: Solution (3/3) — Chart and Interpretation

- Step 3: vol points 4.09 (s.e. 0.27), positive on 86% of days; Step 4: 1990–2007 $t = 12.12$, 2008–2026 $t = 2.03$; difference $-71.0\% ^2$ ($p = 0.08$), in vol points -1.01 ($p = 0.06$)
- **Interpretation:** squaring magnifies the crisis windows (2008, 2020), so the variance premium has heavier tails and a wider interval than the vol-point premium; the premium is positive in both subperiods, but the difference in size is not significant at 5%: the data cannot tell whether it changed

C2 [Proposed]: Critique an AI Answer — the Answer

- A student asks an AI assistant: “Price a European call with 90 calendar days to expiry on a non-dividend-paying stock with $S = 100$, $K = 105$, volatility 20%, continuous rate $r = 4\%$; then give the put from put–call parity”
- The assistant answers with this code:

```
import numpy as np
from scipy.stats import norm
S, K, r, sigma = 100.0, 105.0, 0.04, 20          # sigma = 20%
T = 90 / 252                                     # 90 days
d1 = (np.log(S / K) + (r + 0.5 * sigma**2) * T) / (sigma * np.sqrt(T))
d2 = d1 - sigma * np.sqrt(T)
call = S * norm.cdf(d1) - K * np.exp(-r * T) * norm.cdf(d2)
put = call + S - K * np.exp(-r * T)             # put-call parity
print(f"call_{S}={call:.2f},_{K}put_{S}={put:.2f}")
```

- “The code follows Black–Scholes exactly; the put comes from parity, so no second formula is needed”

C2 [Proposed]: Critique an AI Answer — Task

Solution discussed in class

□ Questions

- ▶ which three errors does the AI answer contain?
- ▶ which check catches each of them?

□ **Inputs:** the code on the previous slide; everything except three lines is correct

□ Tasks

1. Before running the code, estimate what a 90-day call 5% out of the money should cost: 0.2, 2 or 20.
2. For each error, say what is wrong and why, and how you would detect it: a no-arbitrage bound, a unit check, today's primer, or a second method.
3. Correct the three lines, run the code, report C and P , and check put–call parity numerically.
4. Log the three errors in `AI_ERRORS.md` with the fields: what was asked, what the AI answered, what was wrong, how you found out, the correct version.

□ **Report:** the corrected code and output; one sentence on which error still gives a plausible-looking price, and one on why “the put comes from parity” is not a check of the call

□ **Model:** A5, A7 (Black–Scholes step by step)

B6 [Proposed]: Is the VIX an Efficient Forecast? — Task (1/2)

Solution discussed in class

- **Question:** is VIX_t^2 an unbiased and efficient forecast of the variance realised over the next 21 trading days?
- **Inputs**
 - ▶ The daily series of B5: $RV_{t,t+21}$ and VIX_t^2 , 1990–2026; past variance $RV_{t-21,t}$
 - ▶ For step 4, a log HAR-RV [Corsi \(2009\)](#): $\ln RV_{t,t+21}$ on the logs of the 1, 5 and 22-day averages of daily squared returns (annualised, $\%^2$, each floored at 0.01); level forecast $\exp(\widehat{\ln RV} + \hat{s}^2/2)$, $\hat{s}^2$ the residual variance; QLIKE loss $L = RV/\hat{\sigma}^2 - \ln(RV/\hat{\sigma}^2) - 1$ for a forecast $\hat{\sigma}^2$, zero only for a perfect forecast
- **Model:** B5 (Newey–West inference)
- **Notebook:** section B6

B6 [Proposed]: Is the VIX an Efficient Forecast? — Task (2/2)

Solution discussed in class

- **Question:** is VIX_t^2 an unbiased and efficient forecast of the variance realised over the next 21 trading days?
- **Tasks**
 1. Estimate $RV_{t,t+21} = \alpha + \beta VIX_t^2 + \varepsilon_t$ [Mincer & Zarnowitz \(1969\)](#) with Newey–West errors (21 lags); test $\alpha = 0$ and $\beta = 1$ separately and jointly (Wald, 5%), and report $\text{corr}(\hat{\alpha}, \hat{\beta})$.
 2. Add $RV_{t-21,t}$ as a second regressor; then re-estimate in logs, $\ln RV$ on $\ln VIX^2$, and test $\beta = 1$.
 3. Instrument $\ln VIX_t^2$ with $\ln VIX_{t-21}^2$ [Christensen & Prabhala \(1998\)](#) (just-identified GMM, HAC covariance); report the first-stage t and test $\beta = 1$.
 4. From 2000, re-estimate every 21 days on an expanding window of observed targets; compare VIX^2 with HAR-RV by QLIKE loss and a Diebold–Mariano test with HAC (21 lags) [Diebold & Mariano \(1995\)](#), [Patton \(2011\)](#).
- **Report:** all estimates and tests with decisions at 5%; one sentence each
 - ▶ why can both t -statistics be small while the joint test rejects?
 - ▶ what must the instrument satisfy?
 - ▶ does in-sample significance survive out of sample?

Checks and Project Seed

- Check your results
 - ▶ Dividends and the interest rate are in parity; vega and theta carry their units (per vol point, per day)
 - ▶ No naive standard errors on overlapping windows; no risk-neutral probability read as a forecast
 - ▶ A Sharpe ratio of a short-volatility strategy is reported with its skewness and worst month
- Team-project seed: repeat B3, B4 and B8 on a new Deribit snapshot of Ether options and compare with Bitcoin [Pele et al. \(2023\)](#); delta-hedge one-week Bitcoin options on historical paths [Matic, Packham & Härdle \(2023\)](#)

After the Lecture: a Solved Exercise on the Smile

- B3 uses the implied-volatility smile, which [Chapter 12](#) introduces after this seminar; work through it once you have seen the lecture
- What it needs from the lecture
 - ▶ Total implied variance $w(k) = \sigma_{imp}^2(k) \tau$ at log-moneyness $k = \ln(K/F)$, and the five SVI parameters (level, wing slope, tilt, shift, curvature) [Gatheral & Jacquier \(2014\)](#)
 - ▶ The butterfly condition $g(k) \geq 0$: call prices convex in the strike, so the risk-neutral density is nonnegative
- B3 is [Solved]: reproduce the fit, the bootstrap band and $\min g$ from the notebook section B3

B3 [Solved]: An SVI Smile for Bitcoin — Task

▣ Questions

- ▶ does a five-parameter SVI curve describe the Bitcoin smile?
- ▶ how precisely are its at-the-money level and its skew estimated?

▣ Inputs

- ▶ Option snapshot of the lecture; expiry closest to 30 days: 2026-10-30, 38 days, 56 quoted OTM options (puts below the forward, calls above); for each quote $k = \ln(K/F)$ and $w = \sigma_{mark}^2 \tau$, with σ_{mark} the mark IV
- ▶ SVI Gatheral & Jacquier (2014): $w(k) = a + b(\rho(k - m) + \sqrt{(k - m)^2 + s^2})$; the butterfly function $g(k)$ of the lecture: $g \geq 0$ for all k means a nonnegative density

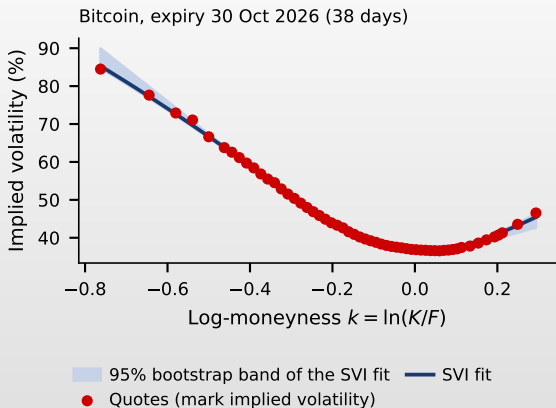
▣ Tasks

1. Fit the five SVI parameters by least squares on w , with $b \geq 0$, $|\rho| < 1$, $s > 0$, and report the root mean squared error in vol points.
2. Resample the quotes with replacement 300 times, refit SVI each time, and report 95% intervals for the ATM volatility and the skew $IV(0.15) - IV(-0.15)$.
3. Evaluate $g(k)$ on $k \in [-1.5, 1.5]$ and report its minimum.

- ▣ **Report:** parameters, intervals, min g and the chart of quotes, fit and band; one sentence each: which parameter controls the skew and its sign; is the skew significantly different from 0; why fit w and not prices

- ▣ Notebook: section B3

B3 [Solved]: Solution (1/2) — Fit and Chart



- Step 1: $\hat{a} = -0.020$, $\hat{b} = 0.102$, $\hat{\rho} = -0.262$, $\hat{m} = -0.079$, $\hat{s} = 0.348$; root mean squared error 0.42 vol points
- Step 2: ATM volatility 36.37%, 95% CI [36.10, 36.87]; skew -2.11 , 95% CI $[-2.73, -1.52]$
- Step 3: $\min_k g(k) = 0.136 > 0$ on $k \in [-1.5, 1.5]$: no butterfly arbitrage on this grid

B3 [Solved]: Solution (2/2) — Interpretation

- ▣ ρ controls the tilt: $\hat{\rho} < 0$ makes the put wing dearer; the skew interval excludes 0, so the tilt is significant at 5%
- ▣ Why w : it is comparable across strikes and nearly linear in the wings, while option prices span four orders of magnitude and would weight the fit towards the dearest options
- ▣ A positive density everywhere also needs $g \geq 0$ outside the grid and the wing conditions of Gatheral & Jacquier (2014)
- ▣ **Interpretation:** the bootstrap resamples smoothed mark volatilities, so the intervals measure fitting noise, not bid–ask uncertainty: they are a lower bound on the true uncertainty

Extension Packs: What Each One Needs

Exercise	Question	Type	Model	Data
A1	are Bitcoin call prices convex in the strike?	Solved	–	option snapshot
A2	which forward and rate does parity imply?	Proposed	A1	option snapshot
A3	what does a dealer earn when $\sigma_{real} \neq \sigma_{imp}$?	Solved	–	–
A4	how fast does the binomial tree converge?	Proposed	A3	–
A6	can one option remove gamma and vega together?	Proposed	A5	–
A8	how large are the errors of a variance strip?	Proposed	A1	–
B2	did delta-hedged option sellers earn a premium?	Proposed	B1, B5	S&P 500, VIX
B4	what is a VIX for Bitcoin?	Proposed	A8, B3	option snapshot
B7	is there a variance risk premium in Bitcoin?	Proposed	B5	DVOL, Bitcoin
B8	how fat are the tails the market prices?	Proposed	B3	option snapshot
C1	is the variance premium a tradable signal?	Proposed	B5, B7	S&P 500, VIX, DVOL
C3	is the Martin bound valid and tight?	Proposed	B5	case-study files

A1 [Solved]: Bounds and Convexity in the Strike — Task

▣ Questions

- ▶ which model-free restrictions must call prices satisfy across strikes?
- ▶ do Bitcoin quotes satisfy them?

▣ Inputs

- ▶ European calls on a non-dividend-paying asset, price $C(K)$, maturity τ , rate r ; no arbitrage, no frictions
- ▶ Bitcoin calls expiring 30 October 2026 in the option snapshot; USD price = BTC premium $\times$ index $S = 85\,265$; forward $F = 85\,742$, so $e^{r\tau} = F/S$; **mark** = the exchange's reference price, **bid/ask** = the best buy and sell quotes

▣ Tasks

1. Prove $\max(S - Ke^{-r\tau}, 0) \leq C \leq S$ by comparing the pay-offs of two portfolios in every state at T .
2. For $K_1 < K_2 < K_3$ with $K_2 = \lambda K_1 + (1 - \lambda)K_3$, prove $\lambda C(K_1) + (1 - \lambda)C(K_3) \geq C(K_2)$.
3. Show that the butterfly $[C(K - h) - 2C(K) + C(K + h)]/h^2$ tends to $e^{-r\tau}q(K)$, with q the risk-neutral density [Breedon & Litzenberger \(1978\)](#).
4. Compute the 84/85/86 thousand butterfly from mark prices, $q(85\,000)$ and $P^Q(84\,000 < S_T < 86\,000)$; then count the negative butterflies among all 732 equally spaced triples, at mark prices and at executable prices (wings at the ask, body at the bid).

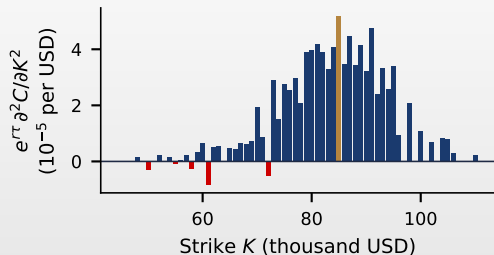
▣ **Report:** the proofs, the butterfly, $q(85\,000)$ and the violation counts; is any violation an arbitrage you could trade?

▣ **Notebook:** section A1

A1 [Solved]: Solution (1/2) — Proofs

- Step 1: a call plus $Ke^{-r\tau}$ in cash is worth $\max(S_T, K) \geq S_T$ at T , so $C + Ke^{-r\tau} \geq S$; the call pays $(S_T - K)^+ \geq 0$, so $C \geq 0$
 - ▶ Upper bound: the share pays $S_T \geq (S_T - K)^+$ in every state, so $C \leq S$
- Step 2: for fixed S_T the map $K \mapsto (S_T - K)^+$ is convex, so $\lambda(S_T - K_1)^+ + (1 - \lambda)(S_T - K_3)^+ \geq (S_T - K_2)^+$ in every state
 - ▶ A portfolio that pays at least as much in every state costs at least as much today
- Step 3: $C(K) = e^{-r\tau} \int_K^\infty (x - K)q(x) dx$, so $\partial_K C = -e^{-r\tau} \int_K^\infty q(x) dx$ and $\partial_K^2 C = e^{-r\tau} q(K)$
 - ▶ The butterfly quotient is the central second difference, so it converges to $e^{-r\tau} q(K)$: convexity is exactly $q \geq 0$

A1 [Solved]: Solution (2/2) — Bitcoin Numbers and Chart



- Adjacent strikes, convex (density ≥ 0)
- Adjacent strikes, not convex at mark prices
- The 84/85/86 thousand butterfly of the task

- Step 4: $C(84\,000) = 4\,957.26$, $C(85\,000) = 4\,414.38$, $C(86\,000) = 3\,922.98$ USD; butterfly = 51.47 USD > 0
- $q(85\,000) \approx 1.0056 \cdot 51.47 / 1\,000^2 = 5.18 \cdot 10^{-5}$ per USD; $P^Q(84\,000 < S_T < 86\,000) \approx 2h q(K) = 10.4\%$
- 7 of 732 mark butterflies are negative (worst -11.80 USD, deep in the money, median spread 255.80 USD); at executable prices: 0
- **Interpretation:** no tradable arbitrage, but a density read from raw marks is negative in places (red bars, 5 of 54 adjacent triples): smooth the smile before differentiating

A2 [Proposed]: Parity as a Regression: the Implied Forward — Task

Solution discussed in class

- ▣ **Question:** which forward price and which USD interest rate do Bitcoin option prices imply through put–call parity?
- ▣ **Inputs:** the expiry of A1; USD mid prices $(bid + ask)/2 \times S$; only strikes where both the call and the put have a bid and an ask; $D = e^{-r\tau}$ and F unknown
- ▣ **Tasks**
 1. Show from put–call parity that $C(K) - P(K) = DF - DK$ for every strike.
 2. Write the OLS regression $C_i - P_i = a + bK_i + e_i$ and derive $\hat{D}$ and $\hat{F}$ from $(\hat{a}, \hat{b})$.
 3. Derive the delta-method standard error of $\hat{F}$ with a heteroskedasticity-robust covariance.
 4. Estimate on the data and compare $\hat{F}$ with the exchange forward, in standard errors.
- ▣ **Report:** $\hat{D}$, $\hat{F}$ with standard errors and the number of strikes
 - ▶ which USD rate does $\hat{D}$ imply?
 - ▶ why might $\hat{F}$ differ from the exchange forward?
- ▣ **Model:** A1 (portfolios with identical pay-offs)
- ▣ **Notebook:** section A2

A3 [Solved]: The Black–Scholes Equation and the Hedging P&L — Task

- **Question:** what does a dealer who sells an option at σ_{imp} and delta-hedges it earn when the price moves with σ_{real} ?
- **Inputs**
 - ▶ $dS_t = \mu S_t dt + \sigma S_t dW_t$, constant r , continuous trading without costs; option value $V(S, t)$ twice differentiable [Black & Scholes \(1973\)](#)
 - ▶ Numbers: the call of A5, sold at $\sigma_{imp} = 25\%$, gamma $\Gamma = 0.0222$ (from A5); 252 trading days a year
- **Tasks**
 1. Replicate V with a self-financing portfolio of Δ shares and cash B at r ; apply Itô's lemma, choose Δ to remove the dW_t term, and derive the Black–Scholes equation from no arbitrage.
 2. For a dealer who sells at σ_{imp} and holds $\Delta = \partial_S V^{imp}$ while S moves with σ_{real} , derive the instantaneous P&L.
 3. Compute today's P&L rate per trading day if $\sigma_{real} = 30\%$, then annualise it with S and Γ frozen.
- **Report:** the equation, the P&L formula and the two numbers
 - ▶ who gains?
 - ▶ why does the P&L depend on the path of S ?
- **Notebook:** section A3

A3 [Solved]: Solution (1/2) — The Equation

- Step 1: Itô: $dV = (V_t + \mu S V_S + \frac{1}{2} \sigma^2 S^2 V_{SS}) dt + \sigma S V_S dW_t$; the self-financing $\Pi = \Delta S + B$, started at $\Pi = V$, has $d\Pi = \Delta dS + rB dt$
- $d(V - \Pi) = (V_t + \frac{1}{2} \sigma^2 S^2 V_{SS} - rB) dt + (V_S - \Delta)(\mu S dt + \sigma S dW_t)$; with $\Delta = V_S$ both the risk and μ disappear
- A riskless position that costs 0 earns 0: $V_t + \frac{1}{2} \sigma^2 S^2 V_{SS} + rSV_S - rV = 0$
- Step 2: the book $-V + \Delta S + (V - \Delta S)$ in cash changes by $-dV + \Delta dS + r(V - \Delta S) dt$ plus interest on the P&L; Itô with σ_{real} and the equation at σ_{imp} give
- $dP\&L = r P\&L dt + \frac{1}{2} \Gamma S^2 (\sigma_{imp}^2 - \sigma_{real}^2) dt$

A3 [Solved]: Solution (2/2) — Numbers

- Step 3: $\frac{1}{2} \cdot 0.02223 \cdot 100^2 \cdot (0.25^2 - 0.30^2)/252 = \frac{1}{2} \cdot 0.02223 \cdot 10^4 \cdot (-0.0275)/252 = -0.0121$ per option and trading day
- Annualised with S and Γ frozen: -3.06 , about 39% of the call price 7.76: a rate, not the realised P&L of this six-month option, since ΓS^2 changes along the path
- **Interpretation:** the seller loses when realised exceeds implied volatility; the loss accrues through Γ , largest near the strike and near expiry, so two paths with the same realised variance give different P&L

A4 [Proposed]: The Binomial Tree Converges to Black–Scholes — Task

Solution discussed in class

▣ Questions

- ▶ how fast does the Cox–Ross–Rubinstein price approach the Black–Scholes price?
- ▶ why does the error oscillate?

- ▣ **Inputs:** CRR tree with N steps of length $\Delta t = T/N$ Cox, Ross & Rubinstein (1979); European call of the lecture:
 $S = K = 100$, $T = 1$, $r = 5\%$, $\sigma = 20\%$

▣ Tasks

1. Show that $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$, $q = (e^{r\Delta t} - d)/(u - d)$ match the risk-neutral mean exactly and the variance of the log step up to $O(\Delta t^2)$.
2. Price the call with $N = 50, 51, 100, 101, 200, 201$ and compute $N(C_N - C_{BS})$ for each.
3. Average the prices for $N = 200$ and $N = 201$ and compare the error with that of $N = 200$.

- ▣ **Report:** a table of C_N and $N(C_N - C_{BS})$; the order of the error, the reason for the even/odd oscillation, and why averaging helps

- ▣ **Model:** A3 (from the tree to the continuous-time equation)

- ▣ **Notebook:** section A4

A6 [Proposed]: A Delta-Gamma Hedge — Task

Solution discussed in class

▣ Questions

- ▶ how many options and shares make a short option book delta- and gamma-neutral?
- ▶ is it then vega-neutral too?

▣ Inputs

- ▶ A dealer has **sold** 1 000 calls of A5 ($S = K = 100$, six months, $r = 3\%$, $\sigma = 25\%$)
- ▶ Hedging instruments: a call with $K = 110$, same expiry, Black–Scholes price 3.90, and the stock; Greeks from the Black–Scholes formulas, gamma per unit of S

▣ Tasks

1. Compute the delta and the gamma of the dealer's position.
2. Find the number n_2 of $K = 110$ calls that makes the gamma zero, then the number of shares that makes the delta zero.
3. Compute the vega of the hedged book and explain the result.

▣ **Report:** a positions table (instrument, quantity, delta, gamma, vega); is the book protected against a change in volatility?

▣ **Model:** A5

▣ **Notebook:** section A6

A8 [Proposed]: The Variance-Swap Strike and Its Estimation Error — Task

Solution discussed in class

- **Question:** how large are the errors of a finite strip of options when it is used to price variance, in a case where the answer is known?
- **Inputs**
 - ▶ Spanning formula of the lecture: $f(S_T) = f(F) + f'(F)(S_T - F) + \int_0^F f''(K)(K - S_T)^+ dK + \int_F^\infty f''(K)(S_T - K)^+ dK$
Bakshi, Kapadia & Madan (2003)
 - ▶ Log-normal benchmark: $F = 100$, $r = 0$, 90 days ($T = 90/365$), flat volatility 20%, so the exact variance is 0.04; OTM prices from Black–Scholes; **discretisation** error = Cboe sum minus the dense integral on $[80, 120]$; **truncation** error = dense integral on $[80, 120]$ minus 0.04
- **Tasks**
 1. Apply the formula with $f(S) = -2 \ln(S/F)$ and show $K_{var} = \frac{2e^{rT}}{T} \left[\int_0^F \frac{P(K)}{K^2} dK + \int_F^\infty \frac{C(K)}{K^2} dK \right]$ for continuous paths.
 2. Compute the Cboe sum with $\Delta K = 5$ on $[80, 120]$ Cboe (2025), the dense integral on $[80, 120]$ and the dense integral on $(0, \infty)$.
 3. Split the total error into discretisation and truncation.
- **Report:** the three variances and volatilities and the two errors
 - ▶ which error dominates here?
 - ▶ what changes for a longer maturity?
- **Model:** A1 (portfolios of options as prices of pay-off shapes)
- **Notebook:** section A8

B2 [Proposed]: Delta-Hedged S&P 500 Options, 1990–2026 — Task

Solution discussed in class

▣ Questions

- ▶ did sellers of delta-hedged one-month S&P 500 calls earn a significant premium?
- ▶ at what tail risk?

▣ Inputs

- ▶ Daily S&P 500 and VIX closes, January 1990 – September 2026; interest and dividends ignored
- ▶ Strategy of the lecture: every 21 trading days sell a one-month ATM call priced with the VIX and hedge it daily at the same volatility; P&L in % of the index [Bakshi & Kapadia \(2003\)](#)

▣ Tasks

1. Compute the mean monthly P&L and its Newey–West standard error with 3 lags, and test H_0 : mean = 0 at 5%.
2. Build a 95% moving-block bootstrap interval for the mean (blocks of 6 months, 2 000 resamples) [Künsch \(1989\)](#).
3. Compute the share of positive months, the skewness, and the share of the total gain lost in the five worst months.

▣ **Report:** the mean with both intervals and the tail statistics; why can a strategy with a high t -statistic still be dangerous?

▣ **Model:** B1 (bootstrap), B5 (Newey–West)

▣ **Notebook:** section B2

B4 [Proposed]: A VIX for Bitcoin — Task

Solution discussed in class

▣ Questions

- ▶ what does the Cboe VIX formula give for Bitcoin?
- ▶ why does it differ from the at-the-money volatilities and from DVOL?

- ▣ **Inputs:** the option snapshot; the two expiries around 30 days (2026-10-09, 2026-10-30); USD mid $Q = \text{BTC mid} \times \text{index}$
 S ; F and $e^{RT} = F/S$ from the Deribit forward

▣ Tasks

1. Select the strip of each expiry with the Cboe rules [Cboe \(2025\)](#): K_0 the first strike at or below F ; puts below K_0 , calls above, the average of both at K_0 ; drop zero bids and stop each wing after two consecutive zero bids.
2. Compute each implied variance $\sigma^2 = \frac{2}{T} \sum_i \frac{\Delta K_i}{K_i^2} e^{RT} Q(K_i) - \frac{1}{T} (F/K_0 - 1)^2$, with ΔK_i half the distance between neighbouring strikes.
3. Interpolate the total variances linearly in time to 30 days and report $100\sigma_{30}$.
4. Compare with the ATM volatilities of the two expiries and with DVOL at the same moment [Alexander & Imeraj \(2020\)](#).

- ▣ **Report:** the two variances, the weight of the near expiry and the index

- ▶ why is the index above both ATM volatilities?
- ▶ which choices could explain the gap to DVOL?

- ▣ **Model:** A8 (strip formula and its errors), B3 (the same snapshot)

- ▣ **Notebook:** section B4

B7 [Proposed]: The Variance Risk Premium of Bitcoin — Task

Solution discussed in class

▣ Questions

- ▶ is the Bitcoin variance risk premium significantly positive?
- ▶ is it larger than the equity premium?

- ▣ **Inputs:** DVOL (Deribit), March 2021 – August 2026, and daily Bitcoin closes (BTC-USD.CC); realised volatility over the next 30 calendar days, $A = 365$

▣ Tasks

1. Compute the mean premium $DVOL_t - \sqrt{RV_{t,t+30}}$ in vol points and its Newey–West standard error with 30 lags, and test it at 5%.
2. Compute the share of positive days and the ratio of mean implied to mean realised volatility, and compare both with the S&P 500 of B5.
3. Count the non-overlapping 30-day periods in the sample.

- ▣ **Report:** premium, test, ratios and effective sample size

- ▶ in which unit is the Bitcoin premium larger?
- ▶ how much can five years of data tell us?

- ▣ **Model:** B5

- ▣ **Notebook:** section B7

B8 [Proposed]: A Risk-Neutral Density with Confidence Bands — Task

Solution discussed in class

▣ Questions

- ▶ how much more probability than the log-normal density does the Bitcoin market put on large moves?
- ▶ how precise is that estimate?

- ▣ **Inputs:** the option snapshot; expiry closest to 90 days: 2026-12-25, 94 days, 58 OTM quotes; SVI fitted as in B3; forward (undiscounted) prices

▣ Tasks

1. Compute the density $q(K) = \partial^2 C / \partial K^2$ from the SVI call prices (Black-76 with F) on a fine strike grid [Breedon & Litzenberger \(1978\)](#).
2. Compute $P^Q(S_T < 0.8F)$ and $P^Q(S_T > 1.2F)$ with 95% pairs-bootstrap intervals (300 refits) and the log-normal values at the ATM volatility.
3. In each refit, recompute the log-normal value at that refit's ATM volatility and give a 95% interval for the difference SVI minus log-normal.
4. Report where the quotes end in K/F and which part of each probability relies on extrapolated wings.

- ▣ **Report:** the two probabilities with intervals vs log-normal

- ▶ is the difference significant?
- ▶ what does the bootstrap band not measure?

- ▣ **Model:** B3 (SVI fit and pairs bootstrap)

- ▣ **Notebook:** section B8

C1 [Proposed]: Is the Variance Risk Premium a Tradable Signal? — Task

Solution discussed in class

▣ Questions

- ▶ would you sell variance on the S&P 500 or on Bitcoin every month (Carr & Wu (2009), Bollerslev, Tauchen & Zhou (2009))?
- ▶ when would you stop?

▣ Inputs

- ▶ A synthetic variance swap with vega notional 1: monthly P&L = $(IV^2 - RV)/(2 IV)$ vol points
- ▶ S&P 500 with the VIX (1990–2026, VIX3M from 2007); Bitcoin with DVOL (2021–2026)

▣ Tasks

1. Compute the monthly P&L series and the Sharpe ratio with a block-bootstrap interval, the skewness, the worst month and the maximum drawdown.
2. Test a timing rule: skip months that start in backwardation (VIX above VIX3M); is it better?
3. Check how fragile the result is to the day on which the monthly grid starts.
4. Compare with the delta-hedged option of the lecture and propose a position size.

▣ **Report:** a table and a chart of cumulative P&L; a verdict with its limitations

▣ **Model:** B5, B7; the predictive regressions of the lecture appendix

C3 [Proposed]: Testing the Martin Bound — Task

Solution discussed in class

- **Question:** is the lower bound on the equity premium of [Martin \(2017\)](#) valid and tight on the course data, as in the case study [Back, Crotty & Kazempour \(2022\)](#)?
- **Inputs**
 - ▶ The authors' daily Martin bound, annualised %, $h = 1, 3, 6, 12$ months, 1990–2020 (columns `1b_m_h` of `market_bounds.csv`, [replication package](#))
 - ▶ S&P 500 (`GSPC.INDX`); daily risk-free rate ([Kenneth French Data Library](#)); monthly Welch–Goyal predictors [Welch & Goyal \(2008\)](#) ([Amit Goyal's data page](#))
- **Tasks**
 1. Compute the daily excess return (price return minus the risk-free rate), compound it over 21, 63, 126 and 252 trading days and annualise with $12/h$, for formation dates 31 January 1990 – December 2020 ($N = 7789$ at 1 month, 7598 at 12 months).
 2. Build the nine positive conditioning variables of Table 3 (carried forward from month-ends) and form the 10 moments $s_{t,T}(1, z_t)$ with $s_{t,T} = R_{t,T}^e - b_{t,T}$ and their mean $\bar{\lambda}$.
 3. Estimate Σ by Hansen–Hodrick [Hansen & Hodrick \(1980\)](#) with $21h$ lags (Newey–West with $1.5 \times 21h$ lags if not positive semi-definite) and compute D_1 , D_2 and their p -values [Kodde & Palm \(1986\)](#).
 4. Regress $R_{t,T}^e$ on a constant and $b_{t,T}$ (Hansen–Hodrick errors) and test $\alpha = 0$, $\beta = 1$ jointly; repeat step 3 with the bound replaced by its time-series mean (Section 4.3).
- **Report:** a table of means, D_1 , D_2 and p -values next to Tables 1 and 3, and one sentence on why the regression test and the tightness test can disagree
- **Model:** B5 (overlapping windows and HAC errors)

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